



2010
TRIAL
HIGHER SCHOOL CERTIFICATE

GIRRAWEE HIGH SCHOOL

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

Attempt Questions 1 – 8
All questions are of equal value

Total marks-120**Attempt Questions 1-8****All questions are of equal value**

Answer each question in a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (15 Marks)

a) $\int \frac{dx}{\sqrt{4x^2 - 9}}$ 2

b) $\int \frac{x dx}{\sqrt{4x^2 - 9}}$ 2

c) (i) Find real numbers a, b and c such that $\frac{4x^2 + 7x + 11}{(x+3)(x^2 + 4)} = \frac{a}{x+3} + \frac{bx+c}{x^2 + 4}$ 2

(ii) Hence show $\int_0^2 \frac{4x^2 + 7x + 11}{(x+3)(x^2 + 4)} = \ln \frac{50}{9} + \frac{\pi}{8}$ 2

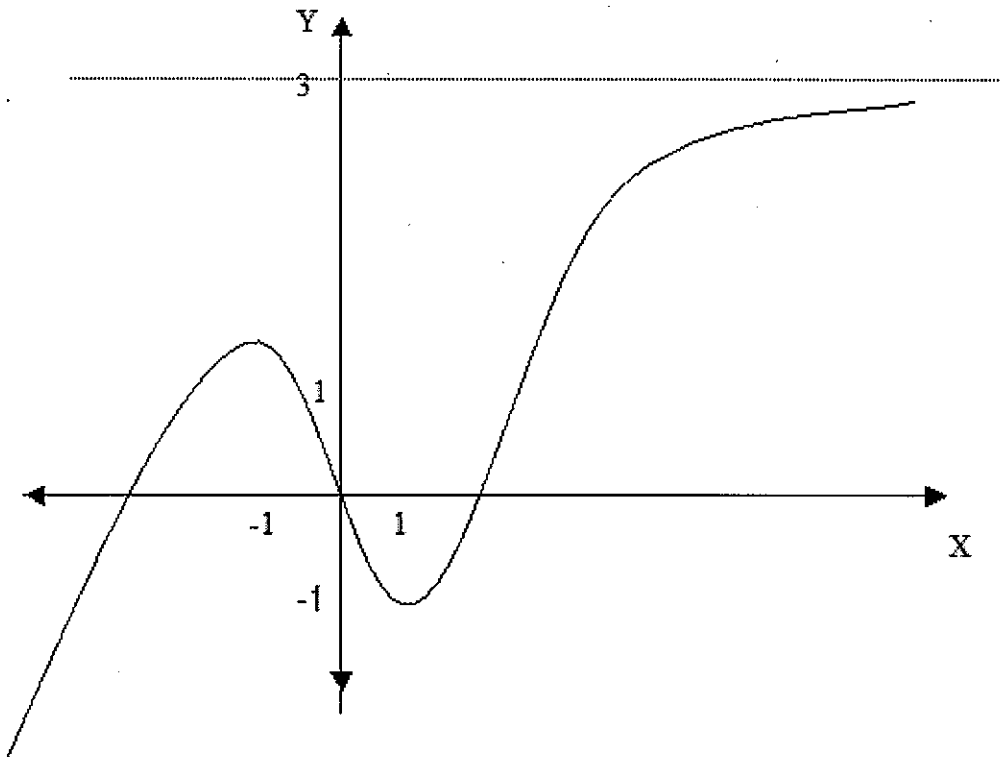
d) Use $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x}$ 2

e) Find $\int \sqrt{x} \ln x dx$ 2

f) Evaluate $\int_2^6 \frac{dx}{x\sqrt{2x-3}}$ 3

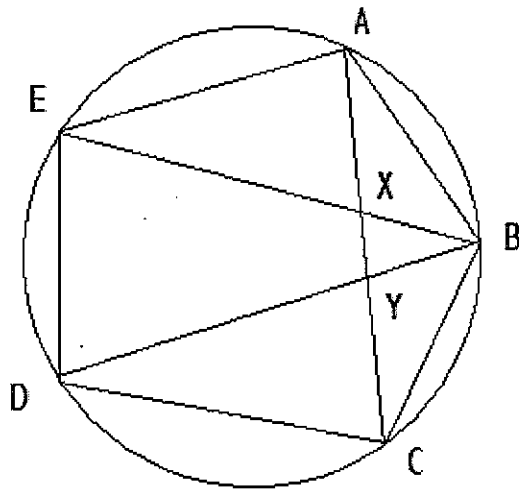
Question 2 (15 marks)

- a) (i) Find real numbers a and b such that $\sqrt{9-40i} = a+ib$ 2
- (ii) Hence find the solutions to $z^2 - 3z + 10i = 0$ 2
- b) (i) Sketch the graph of $|z - 4i| = 2$ 2
- (ii) Hence find the greatest and least values of $\arg z$ 2
- c) If $\omega = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$ where k is an integer and $\omega \neq 1$
- (i) Show that $\omega^n + \omega^{-n} = 2 \cos \frac{2nk\pi}{5}$ 2
- (ii) Show that $\omega^5 = 1$ 1
- (iii) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ 1
- (iv) Hence or otherwise show that $(\omega + \omega^{-1})^2 + (\omega^2 + \omega^{-2})^2 = 3$ 2
- (v) Deduce that $(\cos \frac{2k\pi}{5})^2 + (\cos \frac{4k\pi}{5})^2 = \frac{3}{4}$ 1

Question 3 (15 Marks)

a) The graph of $y = f(x)$ is shown above. It has a local maximum at $x = -1$ and a local minimum at $x = 1$ the curve asymptotes to $y = 3$. Draw neat sketches of the following.

- | | |
|--------------------------------------|---|
| (i) $y = \ln f(x)$ | 2 |
| (ii) $y = e^{f(x)}$ | 2 |
| (iii) $y = f'(x)$ | 2 |
| (iv) $y = f\left(\frac{1}{x}\right)$ | 2 |

Question 3 (continued)

b) The pentagon ABCDE is inscribed inside the circle, with $BA = BC$. The diagonal AC meets the diagonals BE and BD at X and Y respectively.

- (i) Show that $\angle BCA = \angle BEC$ 1
- (ii) Prove that EDYX is a cyclic quadrilateral 3

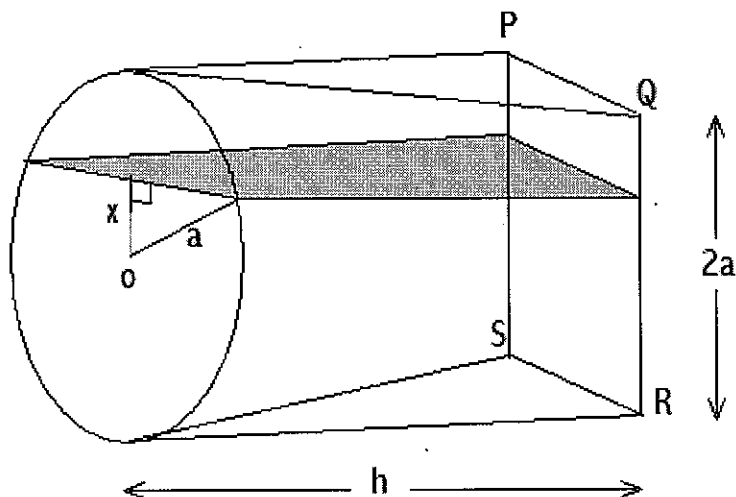
c) The ellipse E has equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$. The point $P(x_1, y_1)$ lies on the ellipse

- (i) Find the equation of the tangent at P to E . 2
- (ii) The chord of contact to the ellipse has equation $\frac{x_0 x}{16} + \frac{y_0 y}{9} = 1$

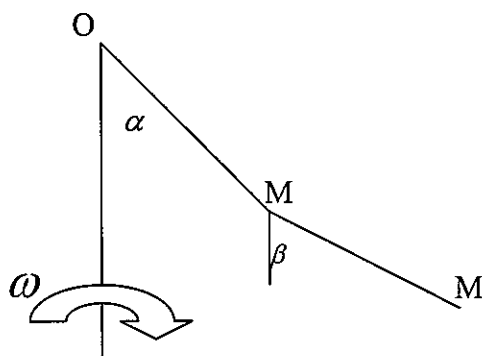
(there is no need to derive this). Show that if the chord passes through the focus, (x_0, y_0) lies on the directrix. 1

Question 4 (15 Marks)

- a) The diagram below shows a solid of length h . It has a circular end of radius a units and the other end is the square PQRS of side $2a$. Horizontal cross-sections parallel to the base of the solid are taken at x as marked on the diagram.



- | | |
|---|---|
| (i) Find the area of the slice at x . | 2 |
| (ii) Express the volume of the solid as an integral | 1 |
| (iii) Find the volume of the solid in (ii) | 3 |
- b) A particle hangs by a light inextensible string of length a from a fixed point O and a second particle of equal mass hangs from the first by an equal string. The whole system moves with constant angular speed ω about the vertical through O, the upper and lower strings making constant angles α and β respectively with the vertical.



- | | |
|--|---|
| (i) Resolve forces vertically and horizontally for both masses m | 4 |
| (ii) Show that $\tan \beta = p(\sin \alpha + \sin \beta)$ | 3 |
| (iii) Show that $\tan \alpha = p(\sin \alpha + 0.5 \sin \beta)$ | 2 |

Where $p = \frac{a\omega^2}{g}$

Question 5 (15 Marks)

a) Given the locus $\frac{x^2}{9-k} + \frac{y^2}{4-k} = 1$, with $k < 4$ Find:

- (i) The eccentricity. 2
- (ii) The coordinates of the foci. 1
- (iii) The equation of the directrices. 1

b) Given the locus $\frac{x^2}{9-k} + \frac{y^2}{4-k} = 1$, with $4 < k < 9$ Find:

- (i) The eccentricity. 2
- (ii) The coordinates of the foci. 1
- (iii) The equation of the directrices. 1

c) A vehicle rounds a banked track of radius 600m, inclined at an angle of θ to the horizontal.

When the car travels at a speed of 10m/s the friction force up the track is equal to the friction force down the track when the vehicle travels at 20m/s. Gravity $g = 9.8\text{m/s}^2$.

- (i) Resolve the forces in mutually perpendicular directions at 10m/s. 2
- (ii) Resolve the forces in mutually perpendicular directions at 20m/s. 2
- (iii) Find the angle θ at which the track is banked. 2
- (iv) Find the speed the car travels to experience no friction force. 1

Hint the mutually perpendicular directions may be parallel and perpendicular to the track or perpendicular and horizontal.

Question 6. (15 Marks)

a) If α, β and γ are the roots of the cubic equation $x^3 - px^2 + qx - r = 0$. Find in terms of p, q and r

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

(iii) $\alpha^3 + \beta^3 + \gamma^3$ 2

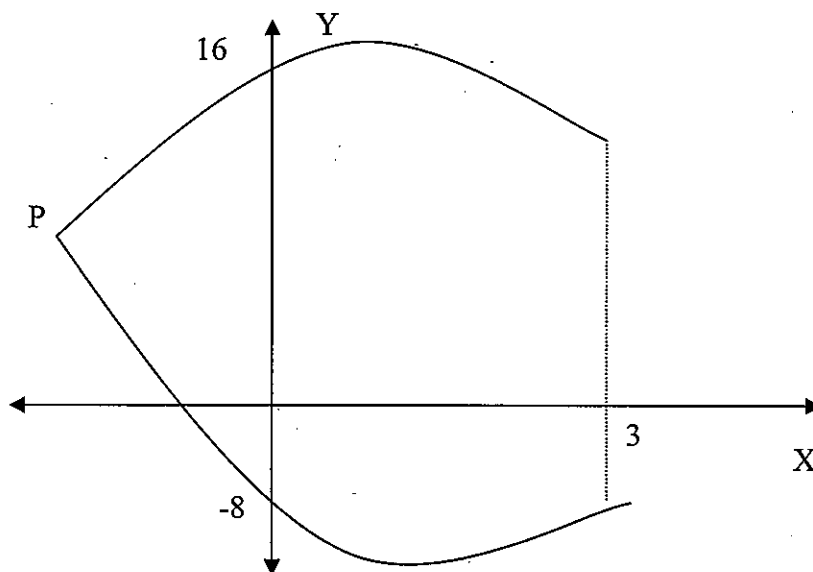
(iv) Hence find a solution to the set of equations 4

$$X + Y + Z = -1$$

$$X^2 + Y^2 + Z^2 = 5$$

$$X^3 + Y^3 + Z^3 = -7$$

b)



The region bounded by the curves $y = 16 - x^2$ and $y = x^2 - 2x - 8$ and the line $x = 3$ is rotated about the line $x = 3$. The point P is the point of intersection of the curves $y = 16 - x^2$ and $y = x^2 - 2x - 8$ in the second quadrant.

(i) Find the coordinates of P. 1

(ii) Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral 3

(iii) Evaluate the integral in (ii) 2

Question 7 (15 Marks)

- a) The sequence of numbers $u_1, u_2, u_3, u_4, \dots, u_n$ is defined as follows

$$u_1 = 1, u_2 = 1 \text{ and } u_n = u_{n-1} + u_{n-2} \text{ for } n > 3$$

Prove that for every positive integer n , $u_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$

Where α and β ($\alpha > \beta$) are the roots of $x^2 - x - 1 = 0$

(Hint in step 1. prove true for $n = 1$ and $n = 2$)

4

b) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

2

(ii) Hence find the value of $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx$

2

c) If n is a positive integer prove $\left(\frac{1+i \tan \theta}{1-i \tan \theta} \right)^n = \frac{1+i \tan n\theta}{1-i \tan n\theta}$

2

- d) The ellipse E has equation $2y^2 - 3xy + 2x^2 = 14$

(i) Using implicit differentiation or otherwise find an expression for the first derivative.

1

(ii) Find the coordinates of any turning points of E

2

(iii) Find the coordinates of any vertical tangents to E

2

Question 8 (15 Marks)

a) (i) Prove that $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$ 2

(ii) Hence find the smallest value of θ such that

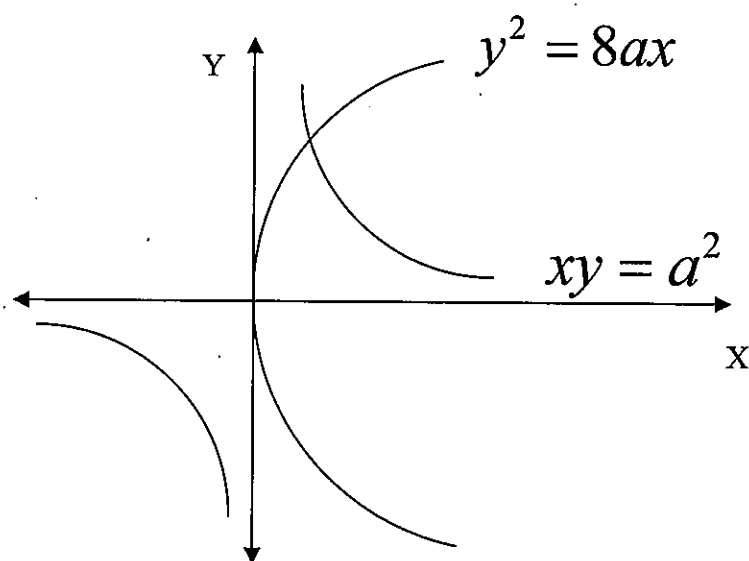
$$(1 + \sin \theta + i \cos \theta)^5 + i(1 + \sin \theta - i \cos \theta)^5 = 0 \quad 2$$

b) Let $I_n = \int_0^1 x(1-x)^n dx \quad n = 0, 1, 2, 3 \dots$

(i) Show that $I_n = \frac{n}{n+2} I_{n-1}$ 3

(ii) Show that $I_n = \frac{1}{2(n+2)C_2}$ 2

c)



Given the hyperbola $xy = a^2$ (H)

and the parabola $y^2 = 8ax$ (P)

(i) Find the coordinates of A the point of intersection of H and P 1

(ii) If $x + y + k = 0$ is the common tangent to H and P, find k 2

(iii) Find the points of contact B on P and C on H 2

(iv) Show that AB is a tangent to H at A 1

SOLUTIONS EXT 2 2010

$$Q1 a) \int \frac{dx}{\sqrt{4x^2-9}} = \frac{1}{2} \int \frac{dx}{\sqrt{x^2-(\frac{3}{2})^2}} \quad (2)$$

$$= \frac{1}{2} \ln \left(x + \sqrt{x^2 - \frac{9}{4}} \right) + C$$

$$b) \int \frac{x dx}{\sqrt{4x^2-9}} = I$$

$$\text{let } u = 4x^2 - 9 \\ du = 8x dx$$

$$I = \frac{1}{8} \int \frac{8x dx}{\sqrt{4x^2-9}}$$

$$= \frac{1}{8} \int \frac{du}{\sqrt{u}}$$

$$= \frac{1}{4} \sqrt{u} + C$$

$$= \frac{1}{4} \sqrt{4x^2-9} + C \quad (2)$$

$$c) (i) \frac{4x^2+7x+11}{(x+3)(x^2+4)} = \frac{a}{x+3} + \frac{bx+c}{x^2+4}$$

$$4x^2+7x+11 = a(x^2+4) + (x+3)(bx+c)$$

$$\text{if } x = -3$$

$$36 - 21 + 11 = 13a$$

$$\boxed{a = 2}$$

$$x = 0$$

$$11 = 8 + 3c$$

$$\boxed{c = 1}$$

$$x = 1$$

$$22 = 10 + 4b + 4$$

$$\boxed{b = 2}$$

$$c) (ii) \int_0^2 \frac{4x^2+7x+11}{(x+3)(x^2+4)} dx$$

$$= \int_0^2 \frac{2}{x+3} + \frac{2x}{x^2+4} + \frac{1}{x^2+4} dx$$

$$= \left[2 \ln(x+3) + \ln(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= 2 \ln 5 + \ln 8 + \frac{\pi}{8} - 2 \ln 3 + \ln 4$$

$$= \ln \frac{25 \times 8}{9 \times 4} + \frac{\pi}{8}$$

$$= \ln \frac{50}{9} + \frac{\pi}{8} \quad (2)$$

$$d) t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (\tan^2 \frac{x}{2} + 1)$$

$$\frac{dx}{dt} = \frac{1}{2} (t^2 + 1)$$

$$\frac{2dt}{1+t^2} = dx$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\int_0^{\pi/2} \frac{dx}{1+\sin x} = \int_0^1 \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2dt}{1+2t+t^2} = \int_0^1 \frac{2dt}{(1+t)^2}$$

$$= \left[\frac{-2}{1+t} \right]_0^1$$

$$= 1 \quad (2)$$

e) $\int \sqrt{x} \ln x \, dx$

let $u = \ln x$ $dv = x^{1/2}$
 $du = \frac{1}{x}$ $v = \frac{2}{3} x^{3/2}$

$\int u \, dv = uv - \int v \, du$ (2)
 $= \frac{2}{3} x^{3/2} \ln x - \frac{2}{3} \int x^{3/2} \cdot \frac{1}{x} \, dx$

$= \frac{2}{3} x \sqrt{x} \ln x - \frac{4}{9} x^{3/2} + C$

f) $\int_2^6 \frac{dx}{x \sqrt{2x-3}} = I$

let $u = (2x-3)^{1/2}$
 $\frac{du}{dx} = \frac{1}{(2x-3)^{1/2}}$ $du = \frac{dx}{\sqrt{2x-3}}$

Now $2x-3 = u^2$
 $2x = u^2 + 3$ $x = \frac{1}{2}(u^2 + 3)$

when $x = 2$ $u = 1$, $x = 6$ $u = 3$.

$I \Rightarrow \int_1^3 \frac{2 \, du}{u^2 + 3}$ (3)

$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} \right]_1^3 = \frac{\pi}{3\sqrt{3}}$

Question 2

a) i) $\sqrt{9-40i} = a+bi$
 $9-40i = a^2 - b^2 + 2abi$
 $9 = a^2 - b^2$
 $-20 = ab$
 $a = \pm 5$ $b = \mp 4$
 $\pm(5-4i)$ (2)

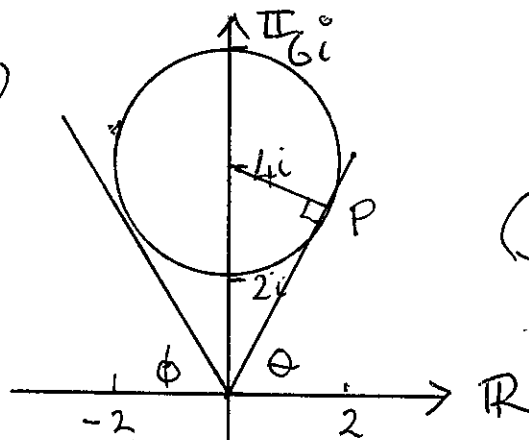
ii) $z^2 - 3z + 10i = 0$

$z = \frac{3 \pm \sqrt{9-40i}}{2}$

$z = \frac{3 \pm (5-4i)}{2}$

$z = 4-2i, -1+2i$ (2)

b) i)



ii) FROM DIAGRAM TANGENT AT P
 $\sin(90-\theta) = \frac{2}{4}$

$90-\theta = 30$

$\theta = 60^\circ$ (2)

$\therefore$ BY SYMMETRY $\phi = 60^\circ$

$\therefore$ MIN ARG = 60° MAX ARG = 120°

c) $w^n + w^{-n}$

DE MOIVRE THEOREM

$= \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$

$+ \cos -\frac{2k\pi}{5} + i \sin -\frac{2k\pi}{5}$

$\cos \theta$ is an ~~odd~~ even function
 $\sin \theta$ is an odd function

$= \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$

$+ \cos \frac{2k\pi}{5} - i \sin \frac{2k\pi}{5}$

$= 2 \cos \frac{2k\pi}{5}$ (2)

Q2 continued

$$\begin{aligned} \text{ii) } \omega^5 &= \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5} \\ &= \cos 2k\pi + i \sin 2k\pi \\ &= 1. \end{aligned} \quad (1)$$

$$\text{iii) } \omega^5 = 1$$

$$\omega^5 - 1 = 0 \quad (1)$$

$$(\omega - 1)(1 + \omega + \omega^2 + \omega^3 + \omega^4) = 0$$

$$\omega \neq 1 \therefore 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0.$$

$$\text{iv) } (\omega + \omega^{-1})^2 + (\omega^2 + \omega^{-2})^2$$

$$= \omega^2 + 2 + \omega^{-2} + \omega^4 + 2 + \omega^{-4}$$

$$= 4 + \omega^{-4} \omega^5 + \omega^2 + \omega^{-2} \omega^5 + \omega^4$$

$$\text{as } \omega^5 = 1$$

$$= 4 + \omega + \omega^2 + \omega^3 + \omega^4$$

$$= 3 + 1 + \omega + \omega^2 + \omega^3 + \omega^4$$

$$= 3 + 0 = 3. \quad (2)$$

$$\text{v) } \omega + \omega^{-1} = 2 \cos \frac{2k\pi}{5}$$

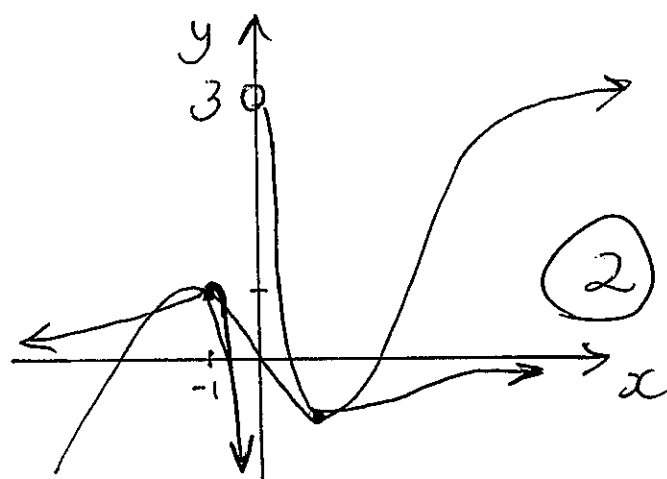
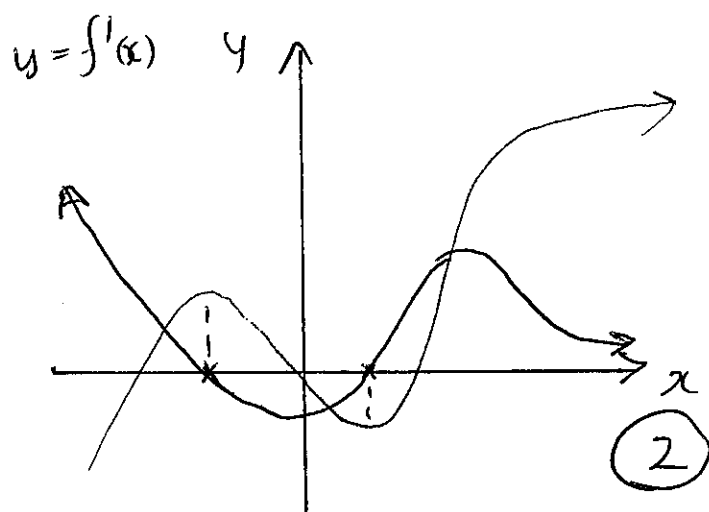
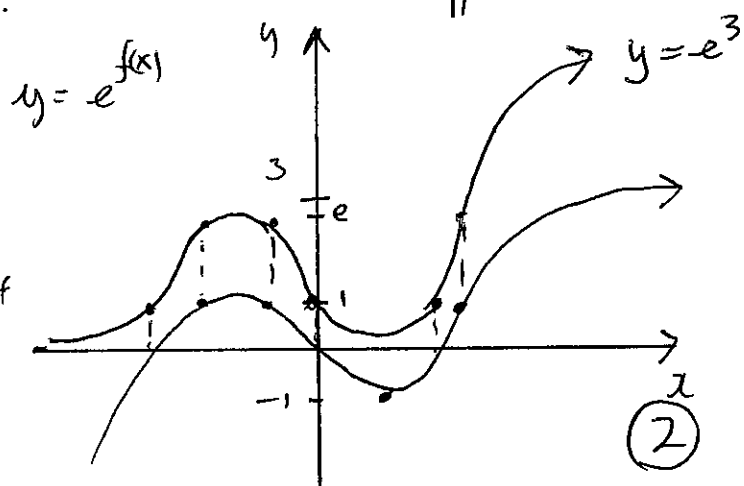
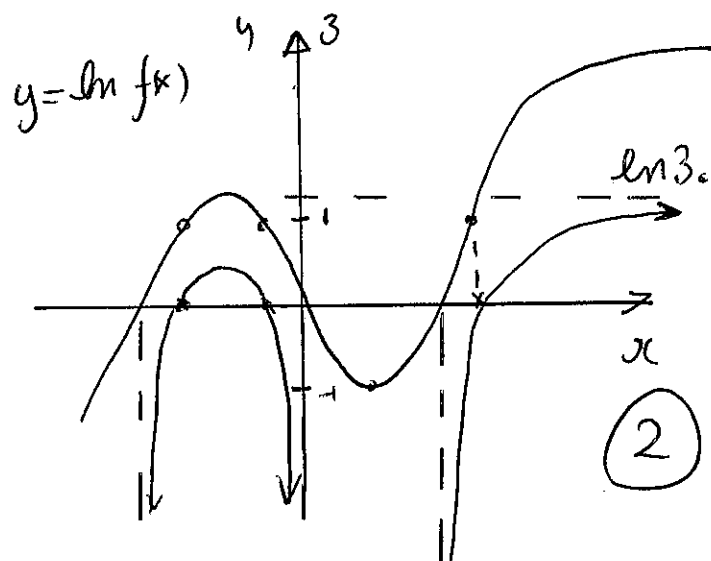
$$\omega^2 + \omega^{-2} = 2 \cos \frac{4k\pi}{5}$$

$$\therefore \left(2 \cos \frac{2k\pi}{5}\right)^2 + \left(2 \cos \frac{4k\pi}{5}\right)^2 = 3 \quad \text{part (iv)}$$

$$\therefore 4 \cos^2 \frac{2k\pi}{5} + 4 \cos^2 \frac{4k\pi}{5} = 3$$

$$\cos^2 \frac{2k\pi}{5} + \cos^2 \frac{4k\pi}{5} = \frac{3}{4} \quad (1)$$

Question 3.



Question 3 (cont.)

$$\angle BCA = \angle BEC \quad \text{Part (i)}$$

$$\angle CBD = \angle CED \quad (\angle's \text{ subtended at the circumference by same arc})$$

$$\therefore \angle BCA + \angle CBD = \angle BEC + \angle CED$$

$$\text{But } \angle AYB = \angle BCA + \angle CBD$$

(EXT $\angle$ IS EQUAL TO SUM INTERIOR OPPOSITE $\angle$ 'S)

$$\therefore \angle AYB = \angle BEC + \angle CED$$

$$\angle AYB = \angle BED \quad (3)$$

(ADDITION OF ADJACENT $\angle$ 'S)

$$\therefore EBYX \text{ IS A CYCLIC QUAD } (\text{INT } \angle \text{ EQUAL EXT OPPOSITE } \angle).$$

$$c) (i) \frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\frac{2x}{16} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{2y}{9} \frac{dy}{dx} = -\frac{2x}{16}$$

$$\frac{dy}{dx} = -\frac{9x}{16y}$$

EQU OF TANGENT

$$y - y_1 = \frac{-9x_1}{16y_1} (x - x_1)$$

$$\frac{y_1 y - y_1^2}{a} = \frac{x_1^2 - x_1 x}{16}$$

$$\frac{x_1 x}{16} + \frac{y_1 y}{9} = \frac{x_1^2}{16} + \frac{y_1^2}{9}$$

But (x_1, y_1) LIES ON E

$$\therefore \frac{x_1 x}{16} + \frac{y_1 y}{9} = 1 \quad (2)$$

(ii) Focus HAS COORDINATES

$$S(ae, 0) = S(\sqrt{7}, 0)$$

$$\frac{x_0 x}{16} + \frac{y_0 y}{9} = 1$$

$$\frac{\sqrt{7} x_0}{16} + \frac{0(y_0)}{9} = 1$$

$$x_0 = \frac{16}{\sqrt{7}}$$

equation of directrix

$$x = \frac{a}{e} = \frac{4}{\frac{\sqrt{7}}{4}} = \frac{16}{\sqrt{7}}$$

(x_0, y_0) LIES ON DIRECTRIX (1)

Question 4.

$$a) 1) \quad \begin{array}{c} \sqrt{a^2 - x^2} \\ \text{ } \end{array} \quad \therefore \text{Length} = 2\sqrt{a^2 - x^2}$$

$$\begin{aligned} \text{Area}_{\text{TRAP}} &= \frac{1}{2}(a+b)h \quad (2) \\ &= \frac{1}{2}(2\sqrt{a^2 - x^2} + 2a)h \\ &= (\sqrt{a^2 - x^2} + a)h \end{aligned}$$

Q4 (a) (cont.)

$$(ii) \delta Vol = (\sqrt{a^2 - x^2} + a) h \delta x$$

$$Vol = \lim_{\delta x \rightarrow 0} 2 \sum_{x=0}^a (\sqrt{a^2 - x^2} + a) h \delta x$$

$$V = 2h \int_0^a \sqrt{a^2 - x^2} + a dx \quad (1)$$

$$(iii) V = 2h \int_0^a \sqrt{a^2 - x^2} dx + 2h \int_0^a a dx$$

Now $\int_0^a \sqrt{a^2 - x^2} dx$ is the $\frac{1}{4}$

$$\text{CIRCLE} = \frac{\pi a^2}{4}$$

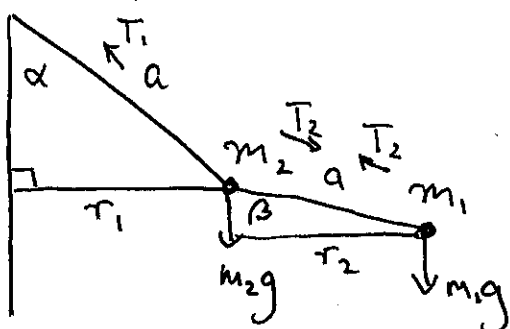
$$V = 2h \frac{\pi a^2}{4} + 2h [ax]_0^a$$

$$V = 2h \frac{\pi a^2}{4} + 2h a^2$$

$$V = 2ha^2 \left(\frac{\pi}{4} + 1 \right) \text{ m}^3 \quad (3)$$

(b) At m_1 (Lower mass)

Vertically



$$T_2 \cos \beta = m_1 g \quad (A) \quad (1)$$

Horizontally

$$T_2 \sin \beta = m_1 r \omega^2 \quad (1)$$

$$= m_1 a (\sin \alpha + \sin \beta) \omega^2 \quad (B)$$

at m_2 (upper mass)

Vertically (1)

$$T_1 \cos \alpha - T_2 \cos \beta = m_2 g \quad (C)$$

HORizontally

$$T_1 \sin \alpha - T_2 \sin \beta = m_2 r \omega^2$$

$$T_1 \sin \alpha - T_2 \sin \beta = m_2 a (\sin \alpha + \sin \beta) \omega^2 \quad (1) \quad (D)$$

$$(ii) \quad (B) \quad \frac{T_2 \sin \beta}{T_2 \cos \beta} = \frac{m_2 a \omega^2 (\sin \alpha + \sin \beta)}{m_1 g}$$

$$\tan \beta = \frac{a \omega^2}{g} (\sin \alpha + \sin \beta)$$

$$\tan \beta = p (\sin \alpha + \sin \beta) \quad (3)$$

(iii) From (A)

$$T_2 \cos \beta = m_1 g$$

$$T_2 = \frac{m_1 g}{\cos \beta}$$

SUBST INTO (C)

$$T_1 \cos \alpha - m_1 g = m_2 g$$

$$T_1 \cos \alpha = 2m_1 g$$

$$T_1 = \frac{2m_1 g}{\cos \alpha}$$

SUBST INTO (D)

$$2m_1 g \tan \alpha - m_1 g \tan \beta = m_2 a \sin \alpha \omega^2$$

$$2 \tan \alpha = \frac{a \omega^2 \sin \alpha}{g} + \tan \beta$$

$$2 \tan \alpha = \frac{a \omega^2 \sin \alpha}{g} + p (\sin \alpha + \sin \beta) \quad (2)$$

$$2 \tan \alpha = 2p \sin \alpha + p \sin \beta \quad (2)$$

$$\tan \alpha = p (\sin \alpha + 0.5 \sin \beta)$$

Question 5

(a) (i) $b^2 = a^2(1 - e^2)$ ellipse

$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$e^2 = \frac{(9-k) - (4-k)}{9-k}$$

$$e^2 = \frac{5}{9-k} \quad e = \frac{\sqrt{5}}{\sqrt{9-k}} \quad (2)$$

(ii) Foci $(\pm ae, 0)$

$$\left(\pm \frac{\sqrt{9-k} \sqrt{5}}{\sqrt{9-k}}, 0 \right) = \pm(\sqrt{5}, 0) \quad (1)$$

(iii) DIRECTRICES $y = \pm \frac{a}{e}$

$$y = \pm \frac{\sqrt{9-k}}{\frac{\sqrt{5}}{\sqrt{9-k}}} = \pm \frac{(9-k)}{\sqrt{5}} \quad (1)$$

(b) (i) $b^2 = a^2(e^2 - 1)$ hyperbola

$$e^2 = \frac{a^2 + b^2}{a^2}$$

$$e^2 = \frac{9-k + (4-k)}{9-k}$$

$$e^2 = \frac{5}{9-k} \quad e = \frac{\sqrt{5}}{\sqrt{9-k}} \quad (2)$$

(ii) Foci $(\pm ae, 0)$

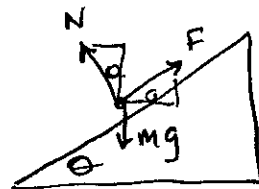
$$= \pm(\sqrt{5}, 0) \quad (1)$$

(iii) DIRECTRICES $y = \pm \frac{a}{e}$

$$y = \pm \frac{9-k}{\sqrt{5}} \quad (1)$$

(c)

(i)



10m/s

Vertically

$$N \cos \theta + F \sin \theta = mg \quad (A)$$

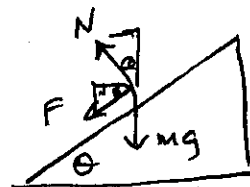
horizontally

$$N \sin \theta - F \cos \theta = \frac{mv^2}{r}$$

$$N \sin \theta - F \cos \theta = \frac{m \cdot 100}{600} \quad (B)$$

(2)

(ii)



Vertically

$$N \cos \theta - F \sin \theta = mg \quad (C)$$

Horizontally

$$N \sin \theta + F \cos \theta = \frac{mv^2}{r} \quad (2)$$

$$N \sin \theta + F \cos \theta = \frac{m \cdot 400}{600} \quad (D)$$

$$(iii) (A) \times \sin \theta - (B) \times \cos \theta$$

$$N \cos \theta \sin \theta + F \sin^2 \theta = mg \sin \theta$$

$$- N \sin \theta \cos \theta - F \cos^2 \theta = \frac{m}{6} \cos \theta$$

$$F = m \left(g \sin \theta - \frac{\cos \theta}{6} \right)$$

$$(D) \times \cos \theta - (C) \times \sin \theta$$

$$N \sin \theta \cos \theta + F \cos^2 \theta = \frac{2m}{3} \cos \theta$$

$$- N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta$$

$$F = m \left(\frac{2 \cos \theta}{3} - g \sin \theta \right)$$

$$\therefore g \sin \theta - \frac{\cos \theta}{6} = \frac{2}{3} \cos \theta - g \sin \theta$$

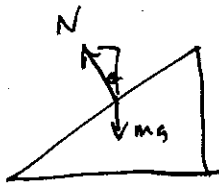
$$2g \sin \theta = \frac{5}{6} \cos \theta$$

$$\tan \theta = \frac{5}{12}$$

$$\theta = 2.435^\circ$$

(2)

(IV) WITHOUT FRICTION



$$N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{r}$$

$$\tan 2.435 = \frac{v^2}{rg}$$

$$v^2 = 250$$

$$v = \sqrt{250} = 15.81 \text{ m/s}$$

(1)

Question 6

$$(a) \quad x^3 - px^2 + qx - r = 0$$

$$(i) \quad \alpha + \beta + \gamma = -\frac{b}{a} = p \quad (1)$$

$$(ii) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= p^2 - 2q \quad (2)$$

$$(iii) \quad \alpha^3 - p\alpha^2 + q\alpha - r = 0$$

$$\alpha^3 = p\alpha^2 - q\alpha + r$$

$$\beta^3 = p\beta^2 - q\beta + r$$

$$\gamma^3 = p\gamma^2 - q\gamma + r$$

$$\alpha^3 + \beta^3 + \gamma^3 = p(\alpha^2 + \beta^2 + \gamma^2) - q(\alpha + \beta + \gamma) + 3r$$

$$= p^3 - 2pq - qp + 3r$$

$$= p^3 - 3pq + 3r \quad (2)$$

$$(iv) \quad \text{let } p = -1$$

$$p^2 - 2q = 5$$

$$p^3 - 3pq + 3r = -7$$

$$\therefore p = -1 \quad q = -2$$

$$\text{and } r = 0$$

x, y, z are the roots of

$$x^3 + x^2 - 2x = 0$$

$$x(x^2 + x - 2) = 0$$

$$x(x+2)(x-1) = 0$$

$$\therefore x, y, z = 0, 1, -2 \quad (4)$$

IN ANY ORDER

$$b) (ii) \quad \delta V = \pi [R^2 - r^2] h$$

$$= \pi [(3-x)^2 - (3-(x+\delta x))^2]$$

$$= \pi [(16-x^2) - (x^2 - 2x - 8)]$$

$$= \pi [9 - 6x + x^2 - (9 - 6x - 6\delta x + x^2 + 2x\delta x + \delta x^2)]$$

$$= \pi [24 + 2x - 2x^2]$$

$$= \pi [6\delta x - 2x\delta x] [24 + 2x - 2x^2]$$

$$= 4\pi [3-x] [12+x-x^2] \delta x$$

$$V = \lim_{\delta x \rightarrow 0} 4\pi \sum_{-3}^3 [3-x] [12+x-x^2] \delta x \quad (3)$$

$$V = 4\pi \int_{-3}^3 (36 - 9x - 4x^2 + x^3) dx$$

$$(i) \quad \text{For } P$$

$$y = 16 - x^2$$

$$y = x^2 - 2x - 8$$

$$16 - x^2 = x^2 - 2x - 8$$

$$0 = 2x^2 - 2x - 24$$

$$0 = 2(x-4)(x+3)$$

$$P(-3, 7) \quad (1)$$

Question 6 (cont)

$$b) (iii) \quad V = 4\pi \int_{-3}^3 (36 - 9x - 4x^2 + x^3) dx$$

$$= 4\pi \left[36x - \frac{9x^2}{2} - \frac{4x^3}{3} + \frac{x^4}{4} \right]_{-3}^3$$

$$= 4\pi \left[\left(108 - \frac{81}{2} - 36 + \frac{81}{4} \right) \right.$$

$$\left. - \left(-108 - \frac{81}{2} + 36 + \frac{81}{4} \right) \right]$$

$$= 4\pi [144] = 576\pi \text{ m}^3 \quad (2)$$

Question 7.

$$a) \quad u_1 = 1 \quad u_2 = 1$$

$$u_n = u_{n-1} + u_{n-2} \quad n > 3$$

$$u_n = \frac{1}{\sqrt{5}} (\alpha^n - \beta^n)$$

where $\alpha > \beta$ roots $x^2 - x - 1 = 0$

$$\alpha, \beta = \frac{1 \pm \sqrt{5}}{2}$$

$$\alpha = \frac{1+\sqrt{5}}{2} \quad \beta = \frac{1-\sqrt{5}}{2}$$

Step 1 Prove true $n = 1, 2$

$$u_1 = 1 \quad u_1 = \frac{1}{\sqrt{5}} (\alpha - \beta)$$

$$u_1 = \frac{1}{\sqrt{5}} (\sqrt{5}) = 1$$

TRUE FOR $n = 1$

$$u_2 = 1 \quad u_2 = \frac{1}{\sqrt{5}} (\alpha^2 - \beta^2)$$

$$= \frac{1}{\sqrt{5}} (\alpha - \beta)(\alpha + \beta)$$

$$= \frac{1}{\sqrt{5}} (\sqrt{5}) (1)$$

$$(\alpha + \beta = -b/a)$$

Step 2. Assume true for $n = k-1, k-2$

$$\text{Now } u_k = \frac{1}{\sqrt{5}} (\alpha^k - \beta^k)$$

$$u_k = u_{k-1} + u_{k-2}$$

$$u_k = \frac{1}{\sqrt{5}} (\alpha^{k-1} + \beta^{k-1} + \alpha^{k-2} + \beta^{k-2})$$

$$= \frac{1}{\sqrt{5}} [\alpha^{k-2}(\alpha+1) + \beta^{k-2}(\beta+1)]$$

$$\text{But } \alpha^2 - \alpha - 1 = 0$$

$$\therefore \alpha^2 = \alpha + 1 \quad \beta^2 = \beta + 1$$

$$= \frac{1}{\sqrt{5}} [\alpha^{k-2}(\alpha^2) + \beta^{k-2}(\beta^2)]$$

$$= \frac{1}{\sqrt{5}} [\alpha^k - \beta^k] \text{ as reqd}$$

Step 3 By the principle of Mathematical Induction true for all n . (4)

$$b) \quad I = \int_0^a f(x) dx$$

$$\text{Let } x = a - u$$

$$\frac{dx}{du} = -1$$

$$\text{when } x = 0 \quad u = a$$

$$\text{when } x = a \quad u = 0$$

$$f(x) = f(a-u)$$

$$I \Rightarrow \int_a^0 f(a-u) (-du)$$

$$= - \int_a^0 f(a-u) du$$

$$= \int_0^a f(a-u) du$$

By Change of Variable

$$= \int_0^a f(a-x) dx \quad (2)$$

$$(11) I = \int_0^{\pi/2} \frac{\sin^3 x}{\cos x + \sin x} dx$$

$$= \int_0^{\pi/2} \frac{\sin^3(\pi/2 - x)}{\cos(\pi/2 - x) + \sin(\pi/2 - x)} dx$$

$$= \int_0^{\pi/2} \frac{\cos^3 x}{\cos x + \sin x} dx$$

$$\therefore 2I = \int_0^{\pi/2} \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x} dx$$

$$I = \frac{1}{2} \int_0^{\pi/2} \frac{\cos x + \sin x (\cos^2 x + \sin^2 x)}{\cos x + \sin x} dx$$

$$I = \frac{1}{2} \int_0^{\pi/2} 1 + \cos x \sin x dx$$

$$= \frac{1}{2} \int_0^{\pi/2} 1 + \frac{1}{2} \sin 2x dx$$

$$= \frac{1}{2} \left[x + \frac{1}{4} \cos 2x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{1}{4} \cos \pi \right) - \left(0 - \frac{1}{4} \cos 0 \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} + \frac{1}{4} + \frac{1}{4} \right]$$

$$= \frac{1}{4} [\pi + 2] \quad (2)$$

$$(c) \left(\frac{1 + i \tan \theta}{1 - i \tan \theta} \right)^n = \left(\frac{\frac{\cos \theta + i \sin \theta}{\cos \theta}}{\frac{\cos \theta - i \sin \theta}{\cos \theta}} \right)^n$$

$$= \left[\frac{\cos \theta + i \sin \theta}{\cos \theta - i \sin \theta} \right]^n$$

DE MOIRE THEOREM

$$= \frac{\cos n\theta + i \sin n\theta}{\cos - n\theta + i \sin - n\theta}$$

$$= \frac{\cos n\theta + i \sin n\theta}{\cos n\theta - i \sin n\theta}$$

$\cos \theta$ is an even fn.

$\sin \theta$ is an odd fn.

$$= \frac{(1 + i \tan n\theta) \cos n\theta}{(1 - i \tan n\theta) \cos n\theta}$$

$$= \frac{1 + i \tan n\theta}{1 - i \tan n\theta} \quad (2)$$

$$d) (i) E \quad 2y^2 - 3xy + 2x^2 = 14$$

$$4y \frac{dy}{dx} - 3y - 3x \frac{dy}{dx} + 4x = 0$$

$$4y - 3x \frac{dy}{dx} = 3y - 4x$$

$$\frac{dy}{dx} = \frac{3y - 4x}{4y - 3x} \quad (1)$$

(ii) FOR TURNING POINTS

$$\frac{dy}{dx} = 0 \quad 0 = 3y - 4x$$

$$y = \frac{4}{3}x$$

$$2\left(\frac{4}{3}x\right)^2 - 3x\left(\frac{4}{3}x\right) + 2x^2 = 14$$

$$\frac{32x^2}{9} - 4x^2 + 2x^2 = 14$$

$$\frac{14x^2}{9} = 14$$

$$x^2 = 9$$

$$x = \pm 3 \quad y = \pm 4 \quad (2)$$

Question 7 (Cont)
(iii) FOR VERTICAL TANGENTS

$$\frac{dy}{dx} \rightarrow 0 \quad : 4y - 3x = 0$$

$$4y = 3x$$

$$y = \frac{3x}{4}$$

$$2\left(\frac{3x}{4}\right)^2 - 3x\left(\frac{3x}{4}\right) + 2x^2 = 14$$

$$\frac{18x^2}{16} - \frac{36x^2}{16} + \frac{32x^2}{16} = 14$$

$$\frac{14x^2}{16} = 14$$

$$x^2 = 16 \quad x = \pm 4 \quad y = \pm 3$$

Question 8.

$$(a) (i) \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \times \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta + i \cos \theta} = -\frac{n}{2} (I_n - I_{n-1})$$

$$= \frac{(1 + \sin \theta)^2 + 2(1 + \sin \theta)i \cos \theta - \cos^2 \theta}{(1 + \sin \theta)^2 + \cos^2 \theta}$$

$$= \frac{1 + 2\sin \theta + \sin^2 \theta + 2(1 + \sin \theta)i \cos \theta - (1 + \sin^2 \theta)}{1 + 2\sin \theta + \sin^2 \theta + \cos^2 \theta}$$

$$= \frac{2\sin \theta + 2\sin^2 \theta + 2(1 + \sin \theta)i \cos \theta}{2 + 2\sin \theta}$$

$$= \frac{2(1 + \sin \theta)\sin \theta + 2(1 + \sin \theta)i \cos \theta}{2(1 + \sin \theta)}$$

$$= \sin \theta + i \cos \theta \quad (2)$$

$$(ii) (1 + \sin \theta + i \cos \theta)^5 + i(1 + \sin \theta - i \cos \theta)^5 = 0$$

$$\therefore \left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^5 + i = 0$$

$$(\sin \theta + i \cos \theta)^5 + i = 0$$

$$[\cos(\pi/2 - \theta) + i \sin(\pi/2 - \theta)]^5 + i = 0$$

$$\cos(5\pi/2 - 5\theta) + i \sin(5\pi/2 - 5\theta) = -i$$

BY DE MOIVRE

$$\cos(\pi/2 - 5\theta) + i \sin(\pi/2 - 5\theta) = -i$$

$$\sin(-5\theta) + i \cos(-5\theta) = -i$$

$$-\sin 5\theta + i \cos 5\theta = -i$$

$$\therefore \sin 5\theta = 0$$

$$\cos 5\theta = -1$$

$$\therefore 5\theta = \pi$$

$$\theta = \pi/5 \quad (2)$$

$$(b) (i) I_n = \int_0^1 x(1-x)^n dx$$

$$\text{let } dv = x \quad u = (1-x)^n$$

$$v = \frac{x^2}{2}$$

$$du = -n(1-x)^{n-1} dx$$

$$I_n = \left[\frac{x^2}{2} (1-x)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot -n(1-x)^{n-1} dx$$

$$= 0 - \frac{n}{2} \int_0^1 [(1-x) - 1] x (1-x)^{n-1} dx$$

$$= -\frac{n}{2} (I_n - I_{n-1})$$

$$\therefore (n+2)I_n = nI_{n-1}$$

$$I_n = \frac{n}{n+2} I_{n-1} \quad (3)$$

$$(ii) I_n = \frac{n}{n+2} I_{n-1}$$

$$I_n = \frac{n}{n+2} \cdot \frac{n-1}{n+1} \cdot \frac{n-2}{n} \cdots \frac{1}{3} I_0$$

$$I_n = \frac{n! 2!}{(n+2)!} I_0$$

$$I_n = \frac{1}{n C_2} I_0$$

$$I_0 = \int_0^1 x dx = \left[\frac{1}{2} x^2 \right]_0^1 = \frac{1}{2}$$

$$\therefore I_n = \frac{1}{2(n C_2)} \quad (2)$$

$$(c) xy = a^2 \quad (H)$$

$$y^2 = 8ax \quad (P)$$

$$\text{From H } y = \frac{a^2}{x}$$

$$\left(\frac{a^2}{x} \right)^2 = 8ax$$

$$\frac{a^4}{x^2} = 8ax$$

Question 8 (cont)

$$\frac{a^3}{8} = x^3$$

$$x = \frac{a}{2} \quad y = 2a.$$

$$A\left(\frac{a}{2}, 2a\right) \quad (1)$$

$$(ii) \quad x + y + k = 0 \quad (T)$$

$$y^2 = 8ax \quad (P)$$

$$y = -(x+k) \quad \text{from (T)}$$

Subst P $(x+k)^2 = 8ax$

$$x^2 + (2k-8a)x + k^2 = 0$$

IF TANG $\Delta = 0$

$$(2k-8a)^2 - 4k^2 = 0$$

$$4k^2 - 32ka + 64a^2 - 4k^2 = 0$$

$$64a^2 = 32ka$$

$$k = 2a$$

TEST WITH (H)

$$x + y + 2a = 0 \quad (T)$$

$$xy = a^2 \quad (H)$$

$$y = -(x+2a)$$

$$-x(x+2a) = a^2$$

$$0 = x^2 - 2ax + a^2$$

$$0 = (x-a)^2 \quad (2)$$

$\therefore$ TANGENT.

(iii) Point of Contact

B on P

$$y^2 = 8ax \quad (P)$$

$$x + y + 2a = 0 \quad (T)$$

$$y = -(x+2a)$$

$$(x+2a)^2 = 8ax$$

$$x^2 + 4ax + 4a^2 = 8ax$$

$$x^2 - 4ax + 4a^2 = 0$$

$$(x-2a)^2 = 0$$

$$x = 2a \quad \therefore y = -4a$$

$$B(2a, -4a) \quad (1)$$

$$C(-a, -a) \quad (1)$$

$$(iv) \quad M_{AB} = \frac{2a+4a}{\frac{a}{2}-2a}$$

$$A\left(\frac{a}{2}, 2a\right) \quad B(2a, -4a)$$

$$= -4.$$

GRADIENT of H at $a/2$

$$\frac{dy}{dx} = -\frac{a^2}{x^2}$$

$$= -\frac{a^2}{a^2/4}$$

$$= -4.$$

$\therefore$ AB IS TANGENT. (1)